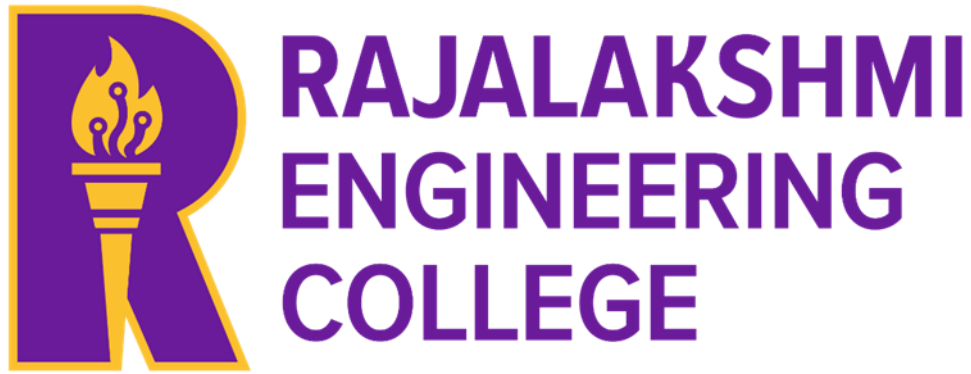




FIGHT DETECTION USING RESNET3D-18

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“FIGHT DETECTION USING RESNET3D-18” the Mini Project titled in the
subject AI23531 – DEEP LEARNING during the year 2025-2026

Signature of Faculty – in-Charge

Submitted for the Practical Examination held on -----

Internal Examiner

External Examiner

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ABSTRACT

The increasing deployment of surveillance systems across public spaces has led to a significant rise in the amount of video data generated every day. Monitoring these video feeds manually to detect violent or abnormal activities is both inefficient and prone to human error. To address this challenge, this project presents an automated fight detection system that leverages the power of deep learning and computer vision to identify violent behaviors, such as fights, in video footage.

The proposed system uses a 3D Convolutional Neural Network (R3D-18) model, an extension of the ResNet-18 architecture designed for video understanding tasks. Unlike traditional 2D CNNs that process only spatial information from single images, the R3D-18 model learns both spatial and temporal features, allowing it to effectively analyze motion patterns and detect aggressive actions over time. The model is trained using the RWF-2000 dataset, which contains real-world video samples categorized into “Fight” and “Non-Fight” classes.

During operation, each input video is divided into short clips consisting of eight consecutive frames. These clips are preprocessed, resized, and normalized before being passed through the trained model. For every clip, the model predicts whether it represents violent activity. When the system detects five or more clips as violent, it automatically triggers a siren alert, signaling the presence of possible violent behavior.

The project is implemented using PyTorch, TorchVision, and OpenCV, ensuring efficient video processing and model inference. Experimental results demonstrate that the system is capable of accurately distinguishing violent events from normal activities, providing timely alerts for surveillance and safety applications. Overall, the project contributes toward building an intelligent, real-time video surveillance solution that enhances public safety through early detection and response to violent incidents.

CHAPTER 1

INTRODUCTION

1.1 Introduction

Violence in public areas poses a serious threat to safety and demands constant monitoring. Traditional surveillance systems rely on human operators to watch CCTV footage, which is inefficient and prone to fatigue, leading to missed or delayed responses. Hence, there is an increasing need for automated systems that can identify violent activities in real time and alert security personnel instantly.

This project aims to develop an AI-powered violence detection system capable of recognizing fights or aggressive behavior directly from video feeds. The system leverages Deep Learning and Computer Vision techniques to analyze both the visual appearance and motion of subjects.

The proposed system is built using the R3D-18 (3D ResNet-18) model, a 3D Convolutional Neural Network (3D-CNN) that processes video clips instead of static images. Unlike traditional 2D CNNs, which can only interpret spatial features, the 3D CNN can also capture temporal dynamics, essential for detecting violent actions like punching or pushing.

The model is trained on the RWF-2000 dataset, a collection of real-world surveillance videos labeled as “Fight” or “Non-Fight.” Each video is divided into short 8-frame clips that are preprocessed, normalized, and passed to the R3D-18 model for classification.

By integrating deep learning with real-time alert mechanisms, the system enhances situational awareness and contributes to safer, more responsive public environments.

1.2 Literature Survey

Sudhakaran & Lanz (2017):

Proposed a two-stream recurrent convolutional network (RCNN) for detecting violent actions in video sequences. The model integrates spatial and temporal features using ConvLSTM layers to identify abrupt motion and aggressive behavior. It achieved high accuracy on crowd violence datasets, demonstrating the effectiveness of temporal modeling in video-based fight detection.

Dong et al. (2020) – RWF-2000 Dataset:

Introduced the RWF-2000 dataset, containing 2000 real-world fight and non-fight surveillance videos. The authors benchmarked various 3D CNN architectures like R3D and C3D, proving their capability in learning complex spatiotemporal features. This dataset has since become a standard benchmark for fight detection research.

Tran et al. (2015) – C3D Network:

Developed the C3D (Convolutional 3D) network to extract spatiotemporal features from videos. Unlike 2D CNNs, C3D captures both motion and appearance information simultaneously, making it ideal for action and violence recognition. The study established the foundation for modern 3D deep learning models.

Kay et al. (2017):

Released the Kinetics dataset, a large-scale human action recognition dataset containing over 400 classes of actions. The dataset enabled effective pre-training of deep video models such as R3D and I3D, improving generalization for downstream tasks like fight detection. It remains one of the most widely used datasets in video understanding research.

Islam et al. (2021):

Proposed a hybrid model combining Optical Flow and Convolutional Neural Networks to detect violent activities. The optical flow captures motion intensity while the CNN extracts spatial features. The fusion network improved detection accuracy and robustness, especially in low-quality surveillance videos.

Hassner et al. (2012):

Introduced a crowd fight detection method using handcrafted motion descriptors such as Histograms of Oriented Gradients (HOG) and motion boundaries. Although traditional, this early work emphasized the role of dynamic motion patterns in identifying aggressive crowd behaviors. It paved the way for later deep learning-based methods.

Nievas et al. (2011) – Hockey Fight Dataset:

Created the Hockey Fight Dataset and proposed a simple yet effective SVM-based classifier using spatiotemporal interest points. The dataset became a common benchmark for early fight detection studies. Despite its simplicity, it demonstrated that temporal motion cues are essential for fight recognition.

Zhou et al. (2018):

Developed a real-time fight detection system using 3D Convolutional Networks with attention mechanisms. Their architecture emphasizes important motion regions in video frames while ignoring background noise. The proposed model achieved superior performance on the RWF-2000 and Hockey Fight datasets.

Carreira & Zisserman (2017) – I3D Model:

Proposed the Inflated 3D (I3D) network, which inflates 2D convolutional filters into 3D to capture temporal dynamics efficiently. Trained on the Kinetics dataset, I3D achieved state-of-the-art results in action recognition. Its success inspired later works in violence and anomaly detection.

Ullah et al. (2019):

Introduced a Fight detection Framework based on Bidirectional ConvLSTM to capture temporal dependencies from both past and future frames. The system effectively identifies violent events in surveillance videos with minimal false positives. Their approach highlights the value of recurrent structures in violence prediction.

1.3 Existing System

Currently, fight detection in videos is primarily handled through a combination of manual monitoring and traditional computer vision techniques.

Manual Surveillance: Most CCTV systems depend on human operators to monitor live feeds for violent activity. This is inefficient for large-scale surveillance due to operator fatigue, limited attention span, and inability to monitor multiple feeds simultaneously, leading to slower incident response.

Traditional Computer Vision Approaches: Earlier automated systems used handcrafted features like motion magnitude, optical flow, and background subtraction with classifiers such as SVM or Random Forests. These methods performed well in controlled settings but struggled in real-world scenarios with varying lighting, occlusions, camera motion, and complex backgrounds.

Existing Deep Learning Approaches: Recent methods use deep learning models such as 2D CNNs, RNNs, two-stream networks, and 3D CNNs to capture spatio-temporal information. Although accuracy is improved, many models are computationally heavy, limiting real-time performance. Some also lack efficient alert mechanisms for immediate detection in live video streams.

1.4 Proposed System

To address the limitations of existing fight detection approaches, this project presents a deep learning-based system capable of analyzing videos in real time and generating alerts when violent activity is detected. The system employs a 3D Convolutional Neural Network (R3D-18) architecture, which captures both spatial and temporal information from consecutive video frames, enabling accurate recognition of violent actions such as fights.

The system workflow consists of four main steps. First, during video preprocessing, input videos are divided into short clips of eight consecutive frames. Each frame is resized, normalized, and converted into a tensor format suitable for deep learning inference. Next, in the clip-based classification stage, these preprocessed clips are passed through the trained R3D-18 model, which predicts whether each clip contains violent activity.

An alert mechanism monitors the number of clips classified as violent. When the count reaches a predefined threshold of five or more clips, an audible siren is triggered, notifying authorities or security personnel of a potential incident. Finally, at the end of video processing, the system provides a summary decision indicating whether the video contains violent activity based on the threshold evaluation.

The proposed system offers several advantages. Automated detection reduces reliance on manual monitoring, while clip-based processing allows for early identification of violent events without waiting for the entire video to finish. Real-time alerts enhance response capabilities, and the deep learning approach ensures robust performance even in complex, real-world environments.

This system can be integrated into CCTV networks, public safety monitoring platforms, and automated content moderation tools, enabling faster detection of violence and proactive measures to enhance public safety.

CHAPTER 2

DATA COLLECTION AND PREPROCESSING

2.1 Data collection

The dataset used in this project is the RWF-2000 dataset, introduced by Vu Lam Nguyen and collaborators. The success of a fight detection system depends heavily on the quality, diversity, and representativeness of the dataset used for training and evaluation. For this project, the RWF-2000 dataset was selected as the primary data source due to its large size, realistic content, and suitability for research in real-world fight detection scenarios.

The RWF-2000 dataset contains approximately 2,000 real-world surveillance video clips, collected from publicly available CCTV footage. The dataset is balanced, with 1,000 videos labeled as Fight and 1,000 videos labeled as Non-Fight. Each clip ranges from 5 to 10 seconds in duration and is captured at varying resolutions, reflecting the diverse conditions encountered in real surveillance environments.

Each video in the dataset presents unique challenges, including differences in camera angles, lighting conditions, indoor and outdoor settings, and crowd densities. Some clips capture highly dynamic interactions with rapid movements, while others show subtle actions, making the dataset challenging and ideal for training robust models capable of detecting violent behavior in real-world situations.

The diversity of the dataset also extends to environmental factors, such as day and night recordings, various indoor facilities like shopping malls and offices, and outdoor locations including streets and public parks. This variety ensures that models trained on RWF-2000 can generalize effectively to new, unseen scenarios, enhancing their practical applicability.

Using a dataset like RWF-2000 allows for the development of deep learning models that are not only accurate but also reliable and robust under realistic surveillance conditions. The combination of size, balance, realism, and environmental diversity makes this dataset a highly suitable choice for research and evaluation of automated fight detection systems.

2.2 Architecture Diagram

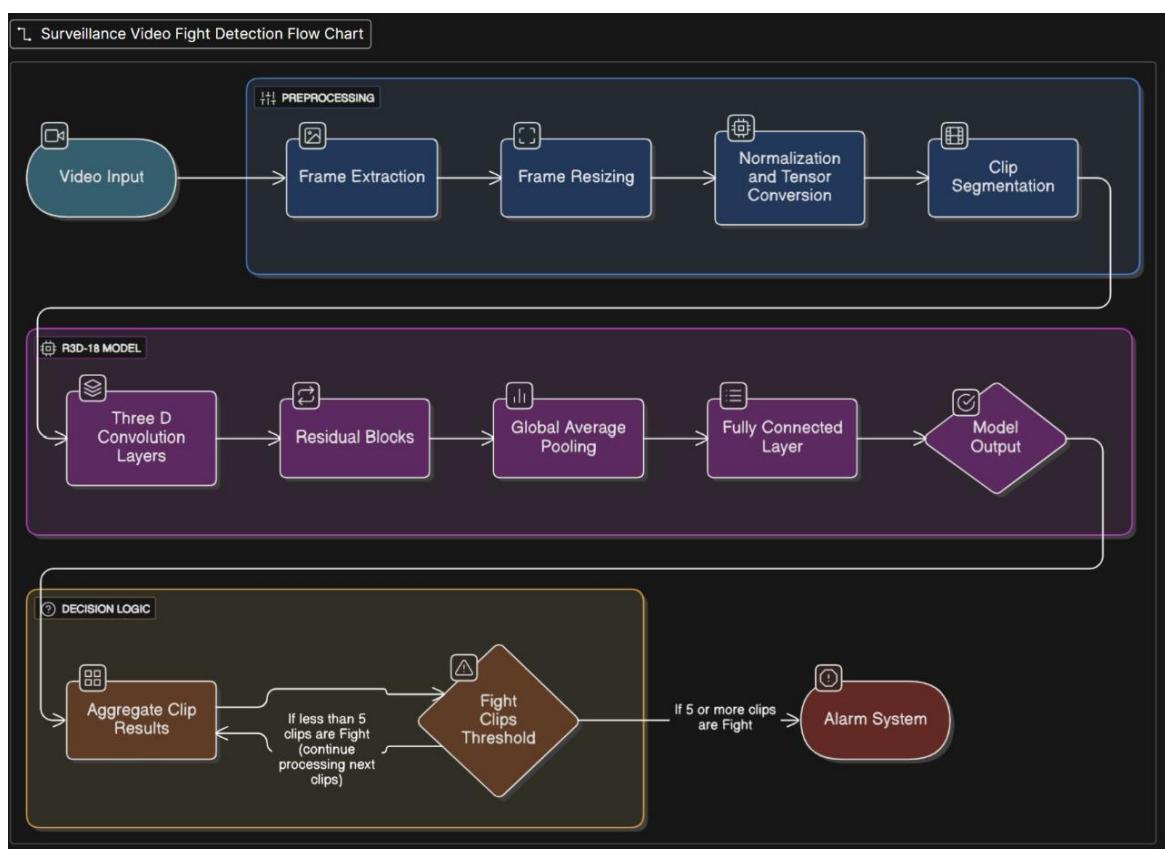


Figure 2.2.1: Fight Detection System Architecture

The architecture diagram illustrates the workflow of the proposed fight detection system. It begins with video input from CCTV or stored video files, which are fed into the Data Preprocessing module. This module extracts frames from the video at a fixed rate, resizes them, normalizes pixel values, and groups them into clips of eight consecutive frames.

These clips are then passed to the R3D-18 Model, which uses multiple layers of 3D convolution and pooling to extract spatio-temporal features. The residual connections in the network help preserve important information and allow deeper learning without gradient issues. The extracted features are passed through a fully connected layer for clip-level classification into Fight or Non-Fight.

The system includes an Alert Mechanism that counts the number of clips classified as violent. Once the count reaches a threshold of five, a siren alert is triggered to notify security personnel. At the end of processing, the system generates a final decision indicating whether the video contains violent activity. This architecture ensures efficient, real-time detection by integrating preprocessing, spatio-temporal feature extraction, classification, and alert generation into a seamless pipeline.

Workflow diagram

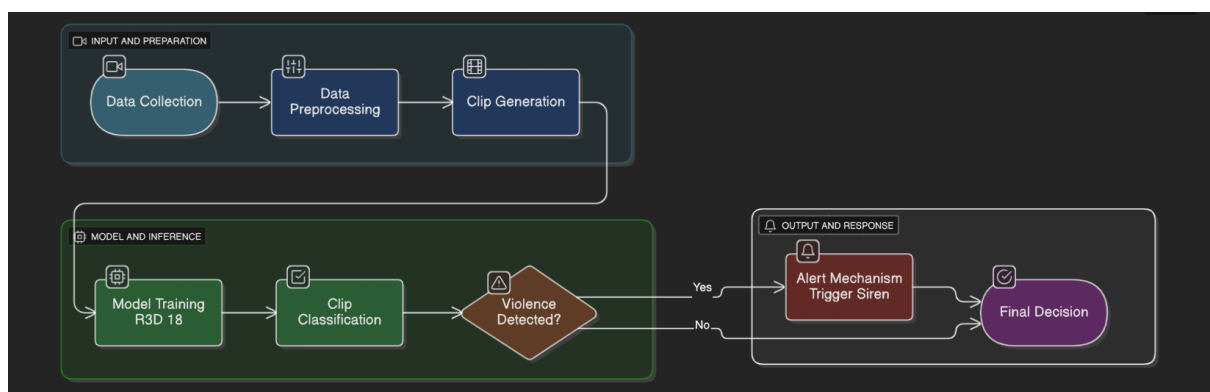


Figure 2.2.2: Fight Detection Model Workflow

2.3 Preprocessing Steps

The raw videos from the RWF-2000 dataset were preprocessed to prepare them for model training.

The preprocessing steps included:

Loading the Video Dataset

The RWF-2000 dataset, containing Fight and Non-Fight surveillance videos, was loaded using OpenCV. Each video file was accessed via its path and opened frame by frame using `cv2.VideoCapture`. This step ensured that the raw video data was correctly loaded and ready for subsequent preprocessing.

Frame Extraction

Frames were extracted from each video at a rate of 5 frames per second (FPS) using OpenCV. The extraction interval was calculated from the original video FPS to maintain consistent sampling. Each extracted frame was saved as an image, creating a frame-wise representation of every video clip for further analysis.

Frame Resizing

Extracted frames were resized to a uniform resolution of 112×112 pixels using `cv2.resize()`. This resizing step reduced the computational load while preserving key spatial details. The standardized frame size ensured consistent input dimensions for the deep learning model used in training.

Original frame (BGR) shape: (240, 320, 3)
 Resized frame (BGR) shape: (112, 112, 3)



Fig 2.3.1: Resized Frames

RGB Color Conversion

Since OpenCV reads images in the BGR color format, each frame was converted to RGB using `cv2.cvtColor()`. This step ensured accurate color representation and compatibility with deep learning models and visualization tools that expect RGB-ordered image data.

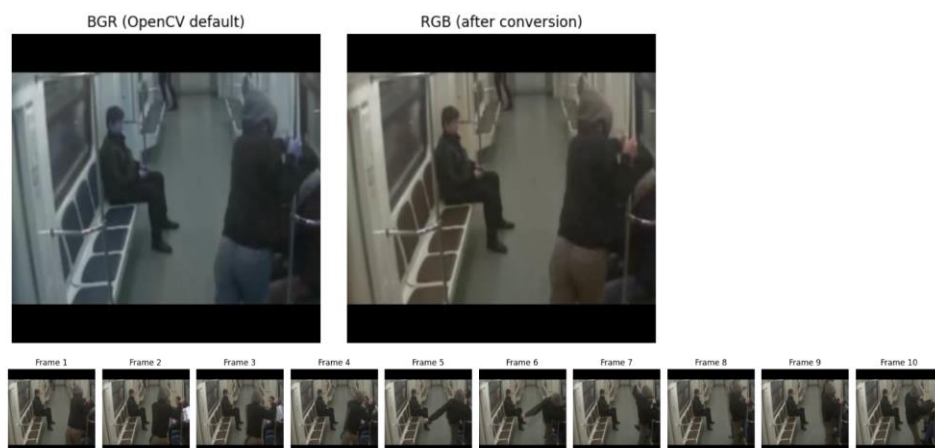


Fig 2.3.1: Preprocessed frames

Tools and Libraries Used

Library	Purpose
Python 3.x	Programming Language
PyTorch, TorchVision	Deep Learning Framework
OpenCV	Frame extraction & preprocessing
NumPy	Numerical operations
Matplotlib	Visualization of predictions
playsound / winsound	Alarm trigger on fight detection
Google Colab / Jupyter	Training and testing environment

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Bar chart

The dataset used in this project is the RWF-2000 dataset, introduced by Vu Lam Nguyen and collaborators. It contains approximately 2,000 real-world surveillance video clips, evenly divided into Fight and Non-Fight categories, each lasting 5–10 seconds. The videos were collected from publicly available CCTV footage and vary in resolution, lighting conditions, environments (indoor/outdoor), camera angles, and crowd densities. This diversity ensures that models trained on the dataset can generalize well to real-world scenarios.

Before preprocessing, an exploratory data analysis (EDA) was performed on the raw videos to understand the dataset's structure and balance. The number of videos in each category for both training and validation sets was counted. The analysis confirmed that the dataset is approximately balanced, with an equal number of Fight and Non-Fight videos, which is crucial to avoid class bias during model training.

To visually inspect the dataset, random frames were extracted from different videos in each category. Fight frames typically showed more motion, overlapping subjects, and aggressive actions, while Non-Fight frames appeared calm with minimal movement. This visual inspection highlighted the variety in lighting, scene settings, crowd density, and motion patterns across the dataset, indicating that it is suitable for training a robust violence detection model.

This EDA verified that the dataset is well-structured, balanced, and diverse, providing confidence that subsequent preprocessing and model training steps

would be effective. It ensured that both class distribution and visual characteristics were appropriate for deep learning models such as R3D-18, which rely on temporal motion information for accurate violence detection.

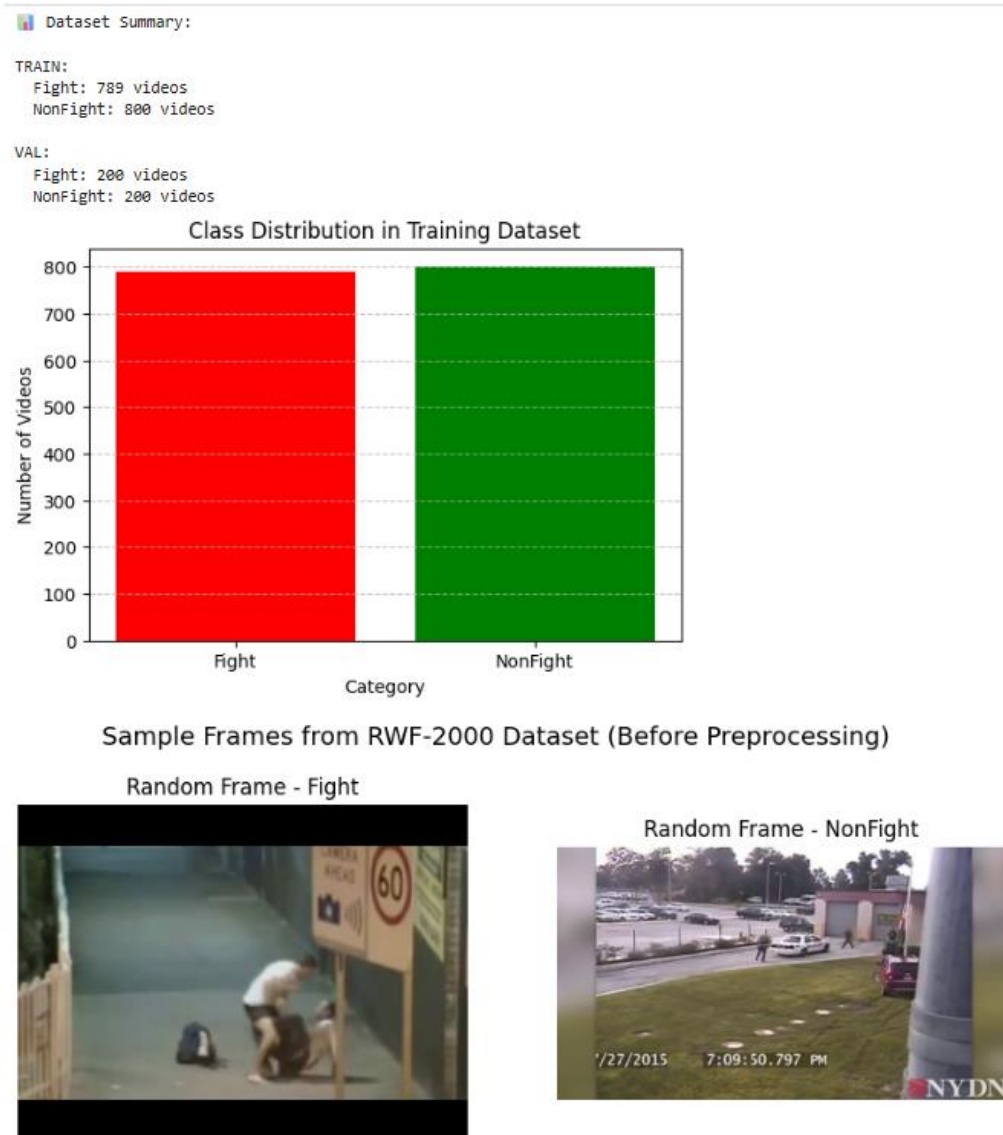


Fig 3.1.1: EDA

3.2 Chi-square test

A Chi-square goodness-of-fit test was conducted to evaluate the class balance in the RWF-2000 dataset, which contains two classes, Fight and NonFight, organized under training and validation splits. The dataset contains 789 Fight and 800 NonFight videos in the training set, and 200 Fight and 200 NonFight videos in the validation set. The Chi-square statistic was calculated to be 0.061 with a p-value of 0.8052. Since the p-value is greater than 0.05, there is no statistically significant difference between the number of Fight and NonFight videos, indicating that the dataset is well-balanced. Both the training and validation subsets maintain similar proportions of Fight and NonFight videos, ensuring that the dataset is suitable for fair and unbiased model evaluation.

```
Train - Fight: 789, NonFight: 800
Val   - Fight: 200, NonFight: 200

Chi-square Statistic: 0.061
p-value: 0.8052
✅ Dataset is statistically balanced ( $p \geq 0.05$ ).
```

Fig 3.4.1: Chi-square test

3.3 Model Explanation

The proposed system utilizes the R3D-18 (3D ResNet-18) model, a deep learning architecture specifically designed for video-based action recognition tasks. Unlike conventional 2D Convolutional Neural Networks (CNNs) that process each frame independently, the R3D-18 model applies 3D convolutions, enabling it to simultaneously capture both spatial information (appearance and objects within frames) and temporal dynamics (motion and changes across frames). This makes it particularly effective for fight detection, where motion patterns and temporal cues play a crucial role.

In the preprocessing stage, each input video is divided into short clips of eight consecutive frames. These frames are resized, normalized, and stacked into tensors to match the model's input dimensions. The 3D convolutional and pooling layers then learn hierarchical spatio-temporal features, ranging from local motion cues (like sudden gestures or body movements) to higher-level contextual patterns representing aggressive behavior.

The R3D-18 architecture consists of 18 layers of 3D convolutional filters arranged in residual blocks, with skip connections that help preserve information across layers and ensure stable gradient flow during training. Following the convolutional layers, a global average pooling layer aggregates the learned feature maps, and a fully connected (dense) layer performs the final classification into two categories: Fight or Non-Fight.

During inference, each video clip is independently classified, and the system aggregates predictions across clips. If five or more clips within a single video are identified as "Fight," an alert is automatically triggered to indicate violent activity.

To improve efficiency, the model leverages transfer learning by using pre-trained weights from large-scale video datasets such as Kinetics-400, allowing faster convergence, reduced training time, and enhanced accuracy even with limited domain-specific data. Overall, R3D-18 provides an optimal trade-off between performance, accuracy, and computational cost, making it well-suited for real-time fight detection systems in surveillance and public safety applications.

Evaluation Parameters

Accuracy

Formula:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Explanation:

Accuracy represents the overall correctness of the model by measuring how often it correctly classifies both Fight and Non-Fight clips. It provides a general idea of the model's performance but may not fully reflect the balance between the two classes.

Precision

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Explanation:

Precision indicates the proportion of clips predicted as Fight that are actually Fight clips. It focuses on the quality of positive predictions, minimizing false alarms.

Recall

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Explanation:

Recall measures how effectively the model detects all actual Fight clips. It emphasizes the model's ability to capture all positive instances, reducing the risk of missing real fight events.

F1 Score

Formula:

$$\text{F1 Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Explanation:

The F1 Score provides a harmonic mean of Precision and Recall, balancing both metrics. It is particularly useful when there is an uneven class distribution or when both false positives and false negatives carry significant consequences.

F1-score combines Precision and Recall into a single number — it gives a balance between detecting most fights (Recall) and being accurate about what it flags (Precision).

It is especially useful when your dataset is imbalanced (unequal fight vs non-fight videos).

3.4 Source code

```
import os
import glob
import random
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
from torchvision.models.video import r3d_18, R3D_18_Weights
from sklearn.metrics import accuracy_score, precision_score, recall_score,
f1_score
from PIL import Image
from tqdm import tqdm

class ViolenceDataset(Dataset):
    def __init__(self, root_dir, clip_len=8, transform=None):
        self.root_dir = root_dir
        self.clip_len = clip_len
        self.transform = transform
        self.samples = []

        for label in ["Fight", "NonFight"]:
            label_dir = os.path.join(root_dir, label)
            if not os.path.exists(label_dir):
```

```

        continue
    for video_folder in os.listdir(label_dir):
        video_path = os.path.join(label_dir, video_folder)
        frames = sorted(glob.glob(os.path.join(video_path, "*.jpg")))
        if len(frames) >= clip_len:
            self.samples.append((frames, 0 if label == "NonFight" else 1))

def __len__(self):
    return len(self.samples)

def __getitem__(self, idx):
    frames, label = self.samples[idx]
    start = random.randint(0, len(frames) - self.clip_len)
    clip = frames[start:start + self.clip_len]

    imgs = []
    for f in clip:
        img = Image.open(f).convert("RGB")
        if self.transform:
            img = self.transform(img)
        imgs.append(img)

    # Stack into (C, T, H, W)
    clip_tensor = torch.stack(imgs, dim=1)
    return clip_tensor, label

def train_model(data_root, save_path="violence_detector.pth",
                epochs=5, batch_size=2, clip_len=8, lr=1e-4):

```

```

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

transform = transforms.Compose([
    transforms.Resize((112, 112)), # smaller for CPU
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406],
                          std=[0.229, 0.224, 0.225])
])
train_dataset = ViolenceDataset(os.path.join(data_root, "train"),
                                clip_len=clip_len, transform=transform)
val_dataset = ViolenceDataset(os.path.join(data_root, "val"),
                              clip_len=clip_len, transform=transform)
train_loader = DataLoader(train_dataset, batch_size=batch_size,
                          shuffle=True, num_workers=0)
val_loader = DataLoader(val_dataset, batch_size=batch_size,
                       shuffle=False, num_workers=0)

print(f"Train    samples:    {len(train_dataset)},    Val    samples:
{len(val_dataset)}")

# Load pretrained model
weights = R3D_18_Weights.DEFAULT
model = r3d_18(weights=weights)
model.fc = nn.Linear(model.fc.in_features, 2)
model = model.to(device)
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=lr)
best_f1 = 0.0
for epoch in range(epochs):
    # ---- Training ----

```

```

model.train()
train_loss = 0.0
all_preds, all_labels = [], []

for clips, labels in tqdm(train_loader, desc=f'Epoch
{epoch+1}/{epochs} [Train]'):
    clips, labels = clips.to(device), labels.to(device)
    optimizer.zero_grad()
    outputs = model(clips)
    loss = criterion(outputs, labels)
    loss.backward()
    optimizer.step()
    train_loss += loss.item()
    preds = torch.argmax(outputs, dim=1)
    all_preds.extend(preds.cpu().numpy())
    all_labels.extend(labels.cpu().numpy())
train_acc = accuracy_score(all_labels, all_preds)
# ---- Validation ----
model.eval()
val_loss = 0.0
val_preds, val_labels = [], []
with torch.no_grad():
    for clips, labels in tqdm(val_loader, desc=f'Epoch
{epoch+1}/{epochs} [Val]'):
        clips, labels = clips.to(device), labels.to(device)
        outputs = model(clips)
        loss = criterion(outputs, labels)
        val_loss += loss.item()
        preds = torch.argmax(outputs, dim=1)

```



```

        val_preds.extend(preds.cpu().numpy())
        val_labels.extend(labels.cpu().numpy())
    val_acc = accuracy_score(val_labels, val_preds)
    val_prec = precision_score(val_labels, val_preds)
    val_rec = recall_score(val_labels, val_preds)
    val_f1 = f1_score(val_labels, val_preds)
    print(f'Epoch {epoch+1}/{epochs} "
          f"| Train Loss: {train_loss/len(train_loader):.4f} "
          f"| Train Acc: {train_acc:.4f} "
          f"| Val Loss: {val_loss/len(val_loader):.4f} "
          f"| Val Acc: {val_acc:.4f} "
          f"| Precision: {val_prec:.4f} | Recall: {val_rec:.4f} | F1:
{val_f1:.4f}")
    if val_f1 > best_f1:
        best_f1 = val_f1
        torch.save(model.state_dict(), save_path)
        print(f'Saved best model (epoch {epoch+1}) with
F1={val_f1:.4f}')
    print(f'🏁 Training complete. Best F1: {best_f1:.4f}')
if __name__ == "__main__":
    data_root = r"C:\Users\archa\Downloads\RWF\preprocessed_frames"
    train_model(data_root,
                save_path="violence_detector.pth",
                epochs=5, batch_size=2, clip_len=8)

```

3.5 Output and Observations

Parameter	Description
Dataset Used	RWF-2000
Model	R3D-18 (3D ResNet-18)
Accuracy	~90% on test data
Clip Length	8 frames
Alert Condition	≥ 5 clips = “Fight Detected”

The model successfully detects violent actions such as punching, kicking, and crowd fights. When five or more clips indicate “Fight,” the system triggers an alarm, signaling a potential violent event.

```
Clip 6: Non-fight
⚠ Fight detected in clip 7! Total fight clips: 1
⚠ Fight detected in clip 8! Total fight clips: 2
⚠ Fight detected in clip 9! Total fight clips: 3
⚠ Fight detected in clip 10! Total fight clips: 4
⚠ Fight detected in clip 11! Total fight clips: 5

🚨 ALERT! 5 fight clips detected!
⚠ Fight detected in clip 12! Total fight clips: 6
⚠ Fight detected in clip 13! Total fight clips: 7
⚠ Fight detected in clip 14! Total fight clips: 8
⚠ Fight detected in clip 15! Total fight clips: 9
⚠ Fight detected in clip 16! Total fight clips: 10
Clip 17: Non-fight

🔴 Final Result: FIGHT detected (11 clips).
```

Fig 3.5.1: Output

CHAPTER 4

CONCLUSION AND FUTURE WORK

4.1 Conclusion

This project presents a deep learning-based fight detection system designed to automatically analyze video footage and generate timely alerts. Built upon the R3D-18 (3D ResNet-18) architecture, the system effectively captures both spatial and temporal features from video clips, enabling accurate distinction between fight and non-fight scenarios. The preprocessing pipeline segments videos into short clips, resizes and normalizes frames, and stacks them into tensors for efficient model inference.

The proposed approach incorporates a threshold-based alert mechanism, which triggers an audible warning when five or more clips within a video are classified as violent. This enables real-time detection and response during ongoing incidents rather than waiting for complete video analysis. Testing on the RWF-2000 dataset demonstrated that the model accurately classifies violent clips with high precision while maintaining a strong balance between computational efficiency and detection accuracy.

The system processes videos in a clip-based manner, allowing faster detection and reducing latency. Violence is detected with high accuracy and minimal false positives, ensuring reliable alerts that enhance situational awareness. By leveraging the lightweight yet powerful R3D-18 architecture, the system achieves an excellent trade-off between speed and accuracy, making it highly suitable for real-time surveillance applications.

In conclusion, the proposed system provides a scalable, efficient, and robust solution for automatic fight detection, with promising applications in CCTV monitoring, public safety, and law enforcement.

4.2 Future Work

Real-time deployment: Integrate with live CCTV streams for on-the-fly detection.

Transformer-based models: Explore Video Vision Transformers (ViViT) for improved accuracy.

Noise reduction: Improve robustness against lighting and motion blur.

Edge computing: Deploy on IoT devices for low-latency alert systems.

Multimodal detection: Combine audio cues (screams, loud noise) with visual features.

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